

ERCOT Grid Observatory

Technical human-review packet

PR stack #33 through #55 | 2026-08-27

READY FOR DETAILED HUMAN REVIEW / CONDITIONAL MERGE CANDIDATE

This is not deployment approval. Nothing was merged or deployed during this remediation. Production, Portainer, production SQLite, collectors, Cloudflare, and DNS were not changed.

Primary remediation: generated generic canonical tile bodies are memory-only. SQLite observations remain authoritative; receiver restart regenerates deterministic bytes and strong ETags. Corrections invalidate intersecting ranges only.

All screenshots in this packet use deterministic synthetic fixtures. No production database, browser session, credential, or source row is embedded.

BLOCKED_EXTERNAL - ACCEPTED SCOPE LIMIT: current four-second ESR

ERCOT discontinued the live feed after 2025-12-05. The reviewed product exposes truthful five-minute system-wide storage context and does not claim SOC, resource identity, high-resolution battery response, or causal attribution.

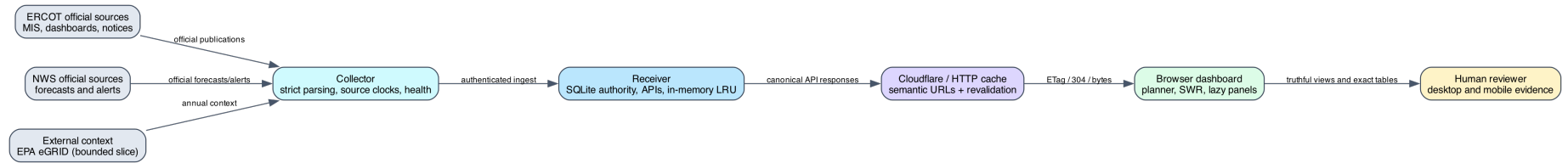
1. Executive review summary

Area	Disposition	Review note
Generic canonical tiles	PASS	No persistent bodies; bounded LRU + singleflight + HTTP/browser caches
Restart behavior	PASS	Fresh process MISS; one generation; identical bytes/ETag; no artifacts
Correction safety	PASS	Affected tile changes; unrelated tile remains byte-identical HIT
Planner reuse	PASS	300 windows; 42.13x aggregate reference reuse; warm SQL = 0
Desktop/mobile review	PASS	14 populated surfaces, each paired desktop + iPhone WebKit
Four-second ESR	ACCEPTED LIMIT	Discontinued external source; no unsupported current claims
Merge/deploy	NOT EXECUTED	Separate human approvals remain required

Reviewers should start with PR #37 and the restart/correction tests, then validate the architecture diagrams, benchmark methodology, data-semantics table, visual pairs, exact-head CI, and stack topology. Review/comment counts are intentionally excluded as evidence.

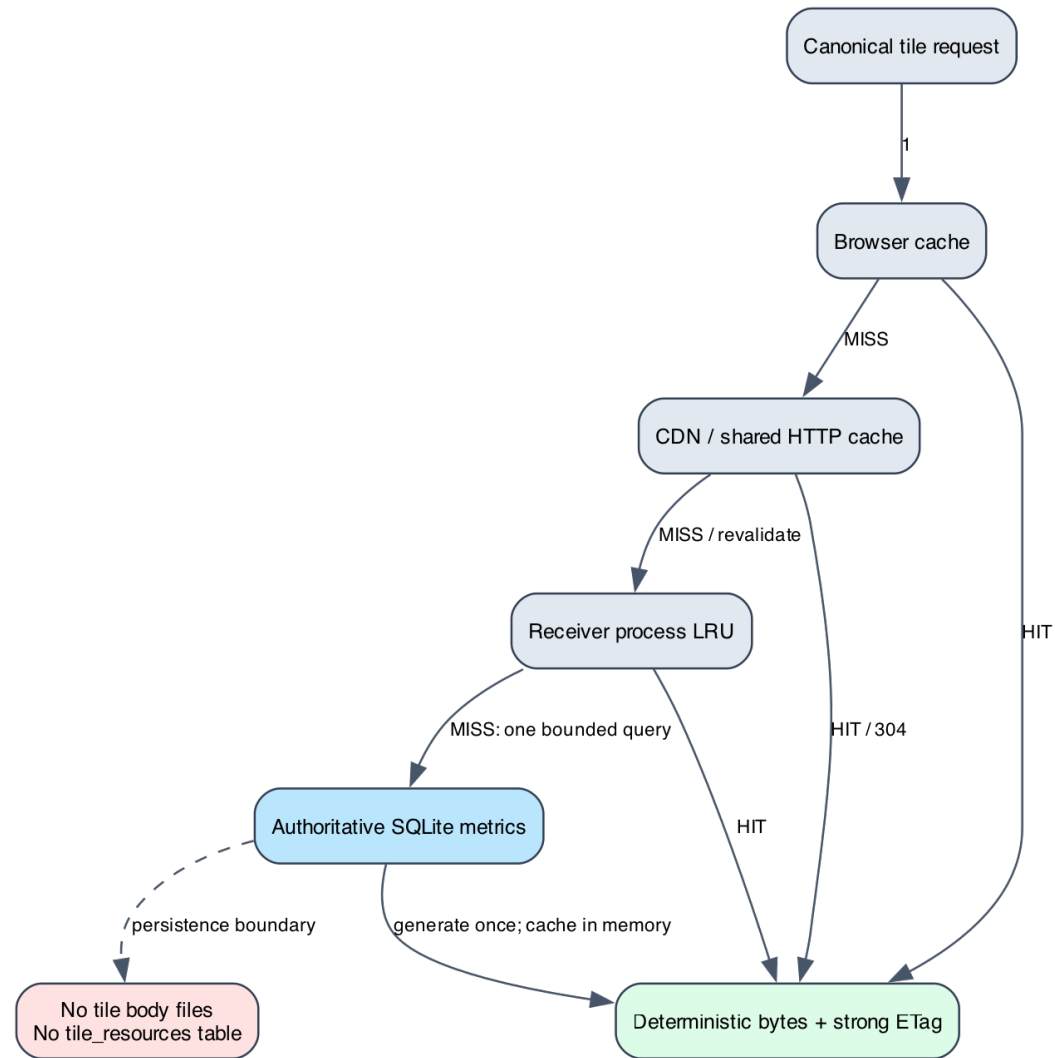
2. System context

Official sources flow through strict collectors to the SQLite-backed receiver, shared HTTP caches, and lazy browser views.



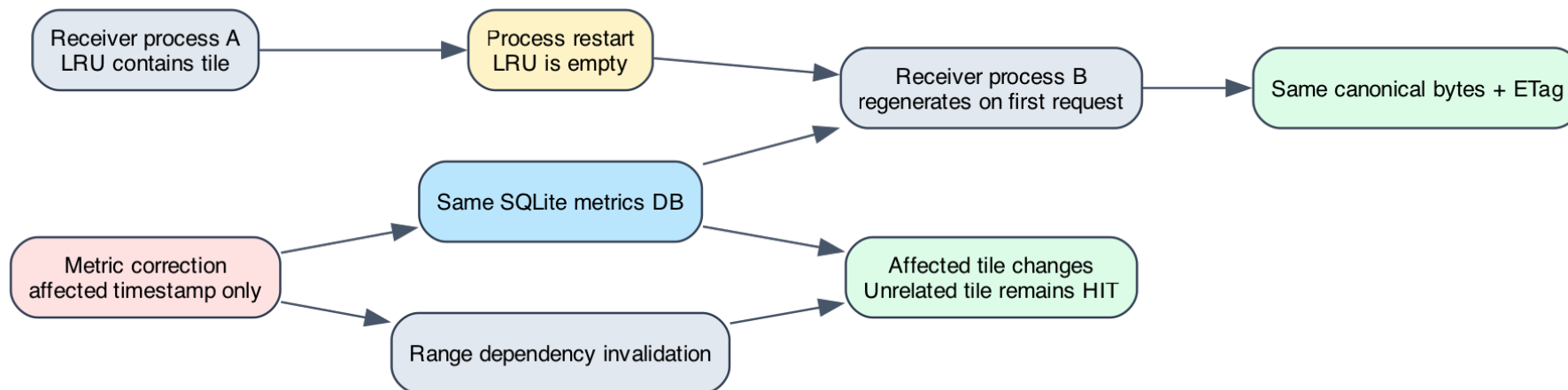
3. Canonical tile request paths

Only observations persist. Generated generic tile bodies live in bounded process memory and downstream caches.



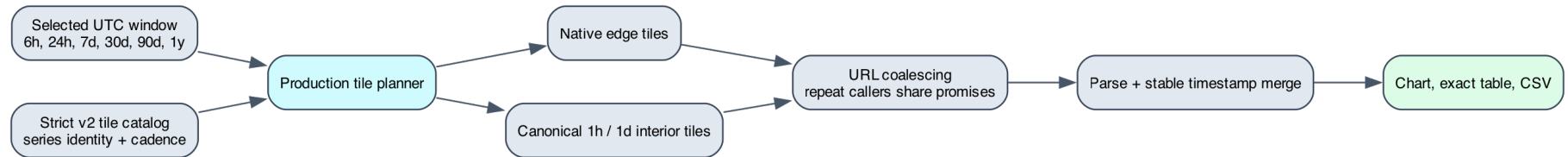
4. Restart and correction

Restart regeneration is deterministic; corrections invalidate only intersecting dependencies.



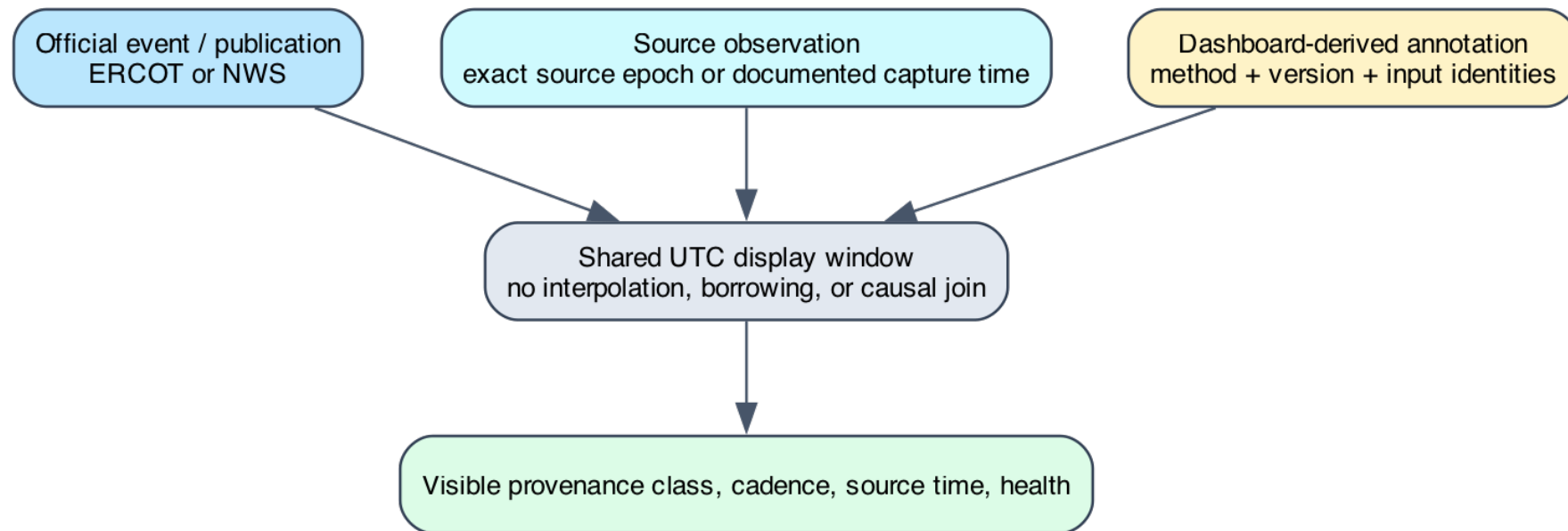
5. Frontend planner

The production planner combines exact native edges and canonical interior tiles without interpolation.



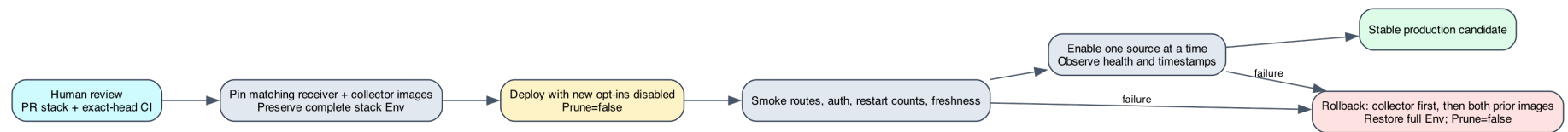
6. Provenance and semantics

Official, source-observation, and derived evidence classes stay distinct.



7. Deployment and rollback

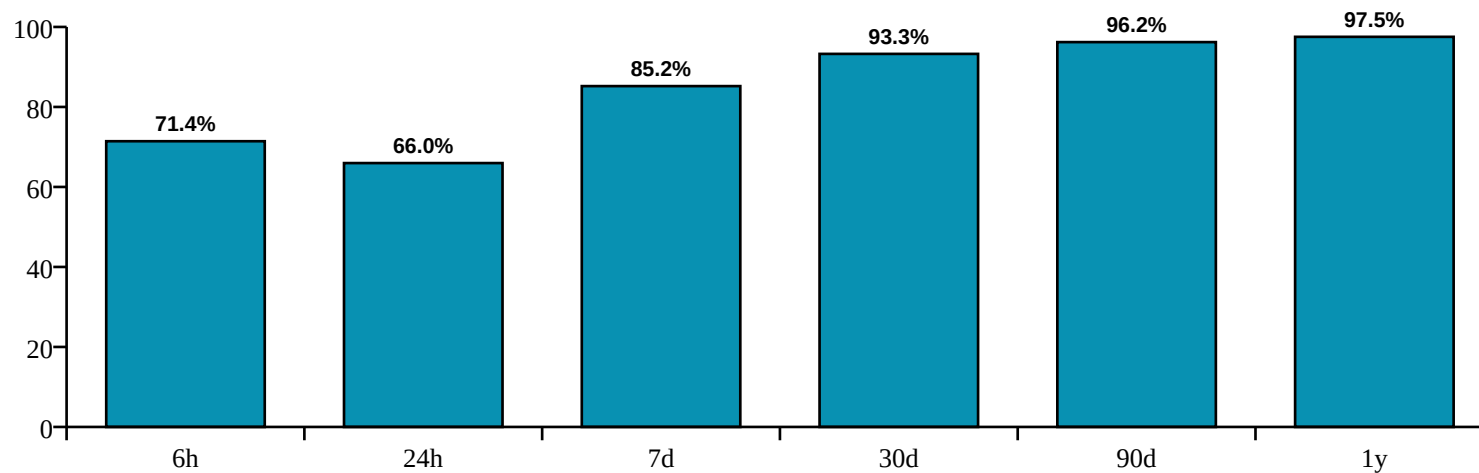
Deployment remains separate and requires pinned images, complete Env preservation, opt-in rollout, and rollback proof.



8. 300-window canonical tile benchmark

Deterministic synthetic fixture; production TypeScript planner; no production mutation.

Application canonical-URL cache hit ratio



Range	Refs	Unique	Reuse	Cold p50/p95/p99 ms	Warm p50/p95/p99 ms	Warm SQL
6h	105	30	3.50x	1.292 / 6.635 / 7.137	0.266 / 1.157 / 1.175	0
24h	238	81	2.94x	1.038 / 5.630 / 5.737	0.277 / 1.088 / 1.130	0
7d	676	100	6.76x	1.039 / 5.646 / 5.832	0.281 / 1.082 / 1.107	0
30d	1,826	123	14.85x	1.054 / 5.469 / 5.611	0.281 / 1.077 / 1.087	0
90d	4,826	183	26.37x	1.801 / 5.434 / 5.664	0.353 / 1.108 / 1.195	0
1y	18,576	459	40.47x	1.857 / 2.259 / 5.583	0.371 / 0.415 / 1.102	0

Honest comparator result: v2 used 623 unique URLs versus 554 for v1, because unaligned windows require exact native edge tiles. The gain is reuse, not lower raw URL cardinality: 26,247 references coalesced to 623 URLs (42.13x), and all warm runs executed zero SQLite statements.

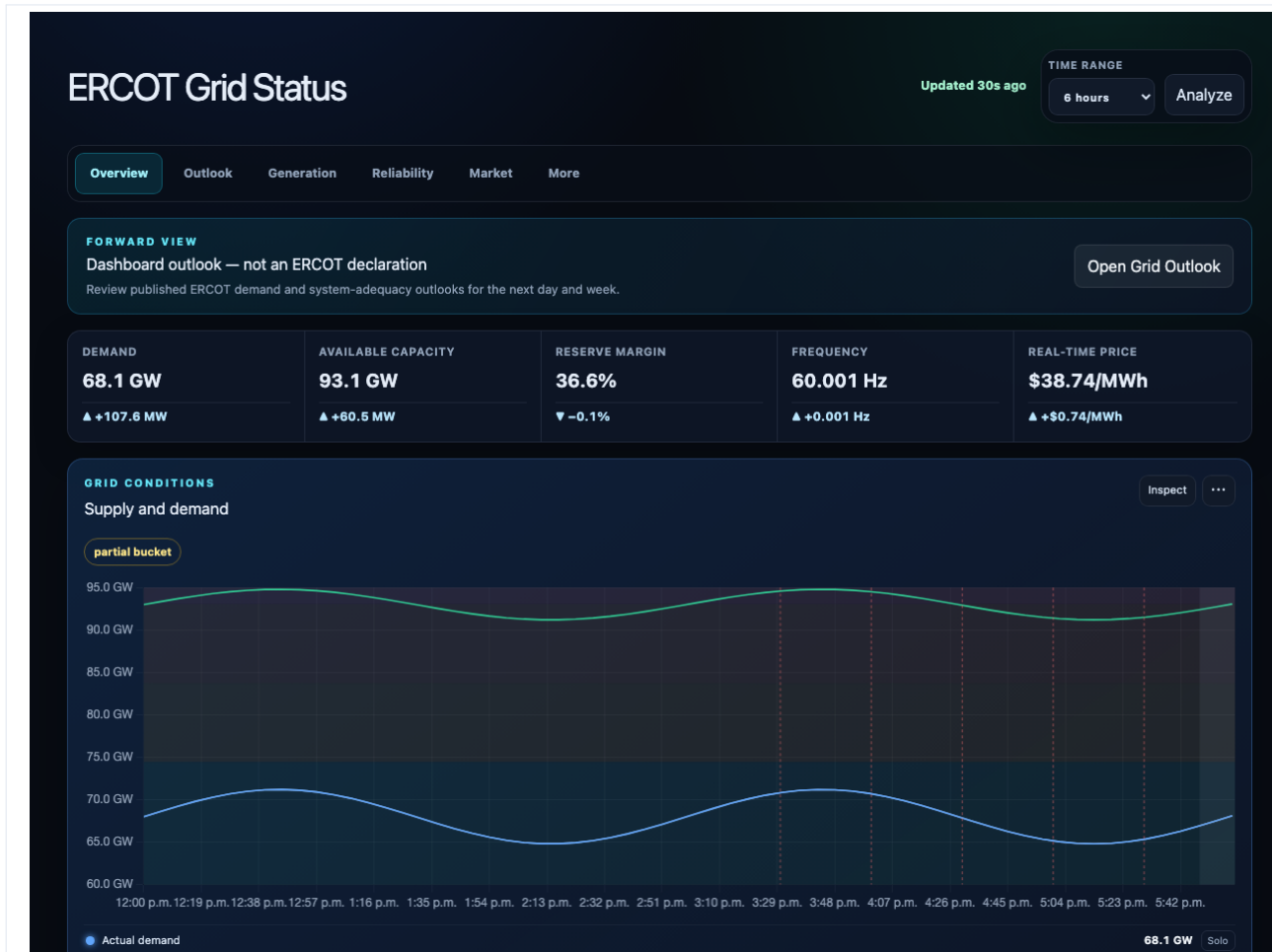
9. Data products and truth boundaries

Surface	Time/source basis	Explicit boundary
Forecast quality	Matched official vintages	Diagnostic pairing; not ERCOT performance declaration
Net load	Demand - wind - solar	Dashboard-derived; no fill, join, or cause
Market mechanics	Exact coherent SCED publications	Context only; no causal decomposition
Congestion geography	LMP/SPP + coincident constraints	No point-price contribution claim
Storage operations	5-minute system-wide aggregate	No SOC, resource identity, intent, or revenue
Storage replay	Shared multi-cadence UTC window	Not high-resolution battery-response attribution
Predictive weather	Four NWS airport points + TX alerts	Not ERCOT zones, statewide extrema, or grid alerts
Unified events	Official/source/derived evidence	Provenance stays separate; TXANS gap visible
Historical context	Observed local-calendar peers	Not official/all-time ERCOT records
Texas Grid	Official planning snapshots	Studied/planned is not installed/committed
External context	Annual EPA eGRID bounded slice	Not live emissions or operational authority

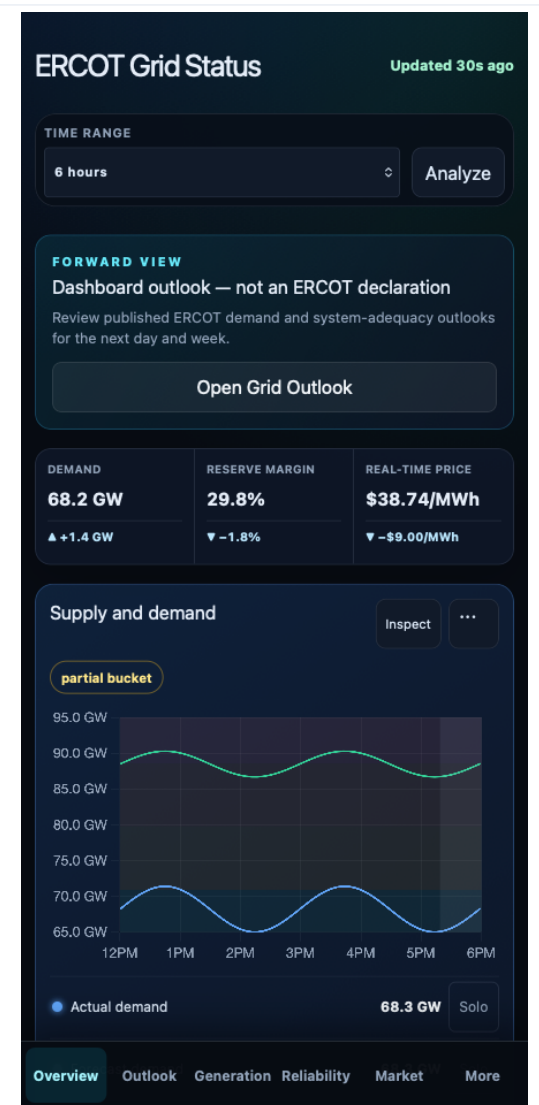
10. Overview

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium



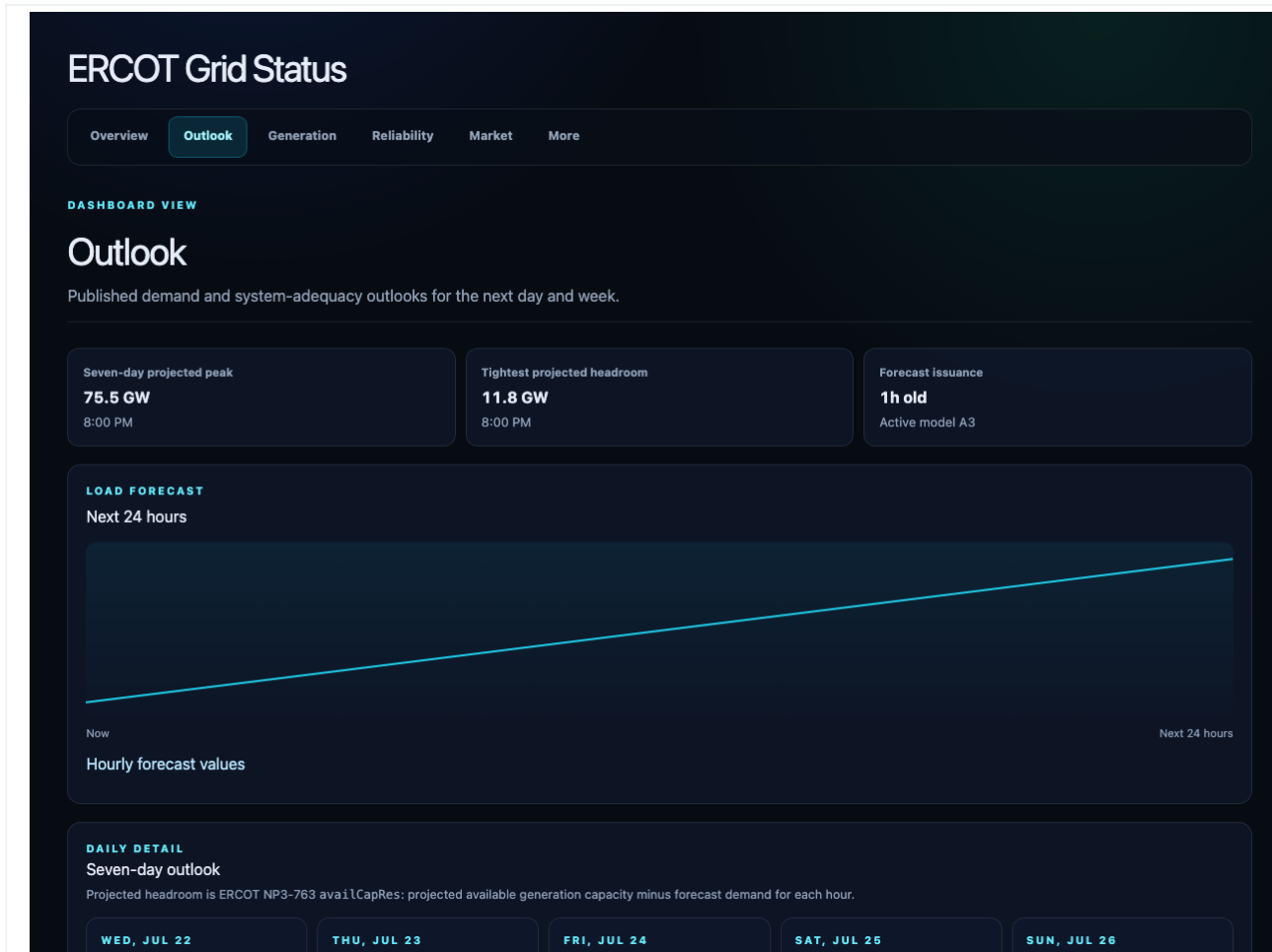
iPhone WebKit



11. Grid Outlook

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium



iPhone WebKit



12. Forecast quality

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

DETERMINISTIC SYNTHETIC REVIEW FIXTURE

HISTORICAL DIAGNOSTICS

Forecast quality

Hide quality details

No history is requested until this panel is opened. Results are diagnostic pairings, not an ERCOT performance declaration.

Forecast series

System load

Wind STWPF

Solar STPPF

Evaluation horizon

1-hour ahead

6-hour ahead

24-hour ahead

Mean absolute error

500.0 MW

Bias

500.0 MW

MAPE

0.7%

Joint sample

24/24 (100.0%)

Signed error = actual – forecast; positive means underforecast. MAE and bias use 24 valid pairs. MAPE uses its own positive-actual sample of 24.

Coverage spans 1 Chicago delivery dates and 1.0 days. Exclusions: none.

Insufficient history for an empirical interval: insufficient_samples, insufficient_delivery_dates, insufficient_sample_span. Requires at least 100 pairs, 30 Chicago delivery dates, 28 days, and 80% joint coverage.

Latest materialized UTC day

Method v1 • Type 7 quantiles

diagnostic product pairing

ercot_public_np3_565_weather_zone_forecast + ercot_public_np6_345_weather_zone_actual_load

iPhone WebKit

DETERMINISTIC SYNTHETIC REVIEW FIXTURE

HISTORICAL DIAGNOSTICS

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Latest materialized UTC day

Overview

Outlook

Generation

Reliability

Market

More

diagnostic product pairing

ercot_public_np3_565_weather_zone_forecast + ercot_public_np6_345_weather_zone_actual_load

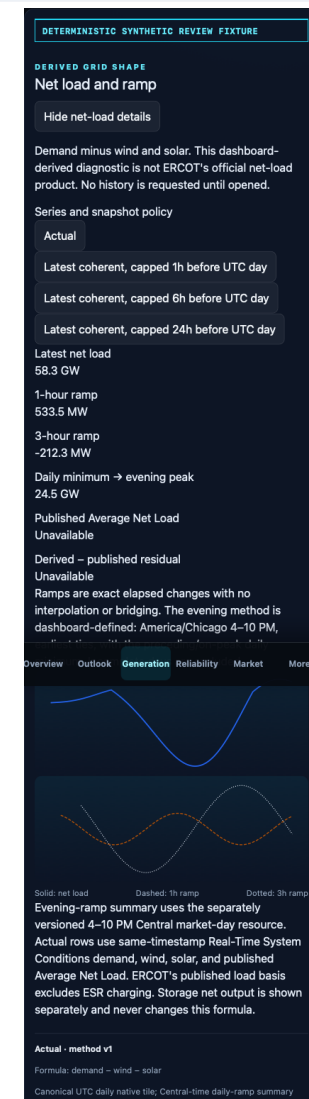
13. Net load and ramp

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium



iPhone WebKit



14. Regional geography

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

SOURCE-NATIVE REGIONAL CONTEXT

Regional load and renewable outlook

Hide regional details

Load, wind, and solar use separate ERCOT source taxonomies. No history is requested until opened.

Regional source layer

Weather-zone loadWind regionsSolar regions

ERCOT region schematic — not geographic boundaries

Some regional sources are stale, failed, or unavailable. Values remain source-labeled.

ercot_public_np6_345_weather_zone_actual_load: healthy; data age 60 seconds; last success 8/17/2026, 7:01:00 PM

ercot_mis_np4_742: stale; data age 10800 seconds; last success 8/17/2026, 7:01:00 PM

ercot_mis_np4_745: healthy; data age 60 seconds; last success 8/17/2026, 7:01:00 PM

Source provenance: source_id ercot_mis_np4_742 · vintage_key rgv1-bbb · issued_at 8/17/2026, 7:00:00 PM · retrieved_at 8/17/2026, 7:01:00 PM

panhandle1.0 GWNo exact prior hour

coastal1.0 GWUp 51.0 MW

south1.0 GWUp 52.0 MW

west1.0 GWUp 53.0 MW

north1.0 GWUp 54.0 MW

Exact wind regions values

Region	Observed	Current	Share	Exact 1h change	Forecast / status
panhandle	8/17/2026, 8:00:00 PM	1.0 GW	10.0%	Unavailable	HSL-potential forecast peak 2.0 GW; forecast error unavailable — generation is curtailment-affected while forecast targets HSL potential
coastal	8/17/2026, 8:00:00 PM	1.0 GW	11.0%	51.0 MW	HSL-potential forecast peak 2.0 GW; forecast error unavailable — generation is curtailment-affected while forecast targets HSL potential
south	8/17/2026, 8:00:00 PM	1.0 GW	12.0%	52.0 MW	HSL-potential forecast peak 2.0 GW; forecast error unavailable — generation is curtailment-affected while forecast targets HSL potential
west	8/17/2026,	1.0 GW	13.0%	53.0 MW	HSL-potential forecast peak 2.0 GW; forecast error unavailable — generation is

Forecast error unavailable: observed generation is curtailment-affected, while the renewable forecast targets HSL potential.

Selected-region hourly generation. Forecast targets HSL potential.

Exact coastal hourly history

Observed interval end	Generation	HSL-potential forecast	Exact 1h change
8/17/2026, 8:00:00 PM	1.2 GW	1.4 GW	100.0 MW
8/17/2026, 9:00:00 PM	1.3 GW	1.5 GW	50.0 MW
8/17/2026, 10:00:00 PM	1.3 GW	1.6 GW	50.0 MW
8/17/2026, 11:00:00 PM	Unavailable	Unavailable	Unavailable
8/18/2026, 12:00:00 AM	1.4 GW	1.6 GW	Unavailable
8/18/2026, 1:00:00 AM	1.4 GW	1.7 GW	50.0 MW

Hourly NP4-742/745 only. Five-minute NP4-743/746 is deliberately deferred.

iPhone WebKit

SOURCE-NATIVE REGIONAL CONTEXT

Regional load and renewable outlook

Hide regional details

Load, wind, and solar use separate ERCOT source taxonomies. No history is requested until opened.

Regional source layer

Weather-zone loadWind regionsSolar regions

ERCOT region schematic — not geographic boundaries

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ercot_mis_np4_745: healthy; data age 60 seconds; last success 8/17/2026, 7:01:00 PM

Source provenance: source_id ercot_public_np6_345_weather_zone_actual_load · observed_at 8/17/2026, 8:00:00 PM · retrieved_at 8/17/2026, 8:01:00 PM

coast1.0 GWNo exact prior hour

east1.0 GWUp 61.0 MW

far-west1.0 GWUp 52.0 MW

north1.0 GWUp 53.0 MW

north-central1.0 GWUp 54.0 MW

south-central1.0 GWUp 55.0 MW

southern1.0 GWUp 56.0 MW

west1.0 GWUp 57.0 MW

Exact weather-zone

Region	Observed	Current	Share
coast	8/17/2026, 8:00:00 PM	1.0 GW	10.0%
east	8/17/2026, 8:00:00 PM	1.0 GW	11.0%
far-west	8/17/2026, 8:00:00 PM	1.0 GW	12.0%
north	8/17/2026,	1.0 GW	13.0%

Load forecast error is a diagnostic NP6-345 actual minus NP3-565 forecast; positive means underforecast. The selected issue is constrained to a 1-2h effective lead and is not an official ERCOT quality score.

Solid: NP6-345 actual. Dashed: coherent NP3-565 forecast.

Exact selected weather-zone history

Interval end	Actual	Forecast	
8/17/2026, 8:00:00 PM	1.2 GW	1.4 GW	-20
8/17/2026, 9:00:00 PM	1.3 GW	1.5 GW	-25
8/17/2026, 10:00:00 PM	1.3 GW	1.6 GW	-25

Hourly NP4-742/745 only. Five-minute NP4-743/746 is deliberately deferred.

15. Market mechanics

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

Energy signal
System Lambda
\$10.00/MWh
+\$1.00/MWh since prior coherent SCED (300s)
8/18/2026, 12:59:00 PM · NP6-322-CD

iPhone WebKit

Energy signal
System Lambda
\$10.00/MWh
+\$1.00/MWh since prior coherent SCED (300s)
8/18/2026, 12:59:00 PM · NP6-322-CD

16. Congestion and price geography

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

15-minute settlement-price matrix
Interval ending 8/20/2026, 12:15:00 PM. Tiles are a schematic list, not Texas coordinates or boundaries.
Fixed meaning: lower prices are blue, near-zero prices neutral, and higher prices orange. Every tile also prints the exact signed value.

Houston HU · HB_HOUSTON -\$42.16/MWh	North HU · HB_NORTH \$18.00/MWh	Pan HU · HB_PAN \$35.00/MWh	South HU · HB_SOUTH \$62.00/MWh	West HU · HB_WEST \$145.00/MWh	Aen LZ · LZ_AEN \$28.00/MWh	Cps LZ · LZ_CPS \$31.00/MWh
Houston LZ · LZ_HOUSTON \$58.00/MWh	Lcra LZ · LZ_LCRA \$44.00/MWh	North LZ · LZ_NORTH \$92.00/MWh	Raybn LZ · LZ_RAYBN \$38.00/MWh	South LZ · LZ_SOUTH \$73.00/MWh	West LZ · LZ_WEST \$225.00/MWh	Busavg SH · HB_BUSAVG \$51.00/MWh
Hubavg AH · HB_HUBAVG \$64.00/MWh						
Settlement point	Type	Price	Interval end			
HB_HOUSTON	HU	-\$42.16/MWh	8/20/2026, 12:15:00 PM			
HB_NORTH	HU	\$18.00/MWh	8/20/2026, 12:15:00 PM			
HB_PAN	HU	\$35.00/MWh	8/20/2026, 12:15:00 PM			
HB_SOUTH	HU	\$62.00/MWh	8/20/2026, 12:15:00 PM			
HB_WEST	HU	\$145.00/MWh	8/20/2026, 12:15:00 PM			
LZ_AEN	LZ	\$28.00/MWh	8/20/2026, 12:15:00 PM			

iPhone WebKit

15-minute settlement-price matrix

Interval ending 8/20/2026, 12:15:00 PM. Tiles are a schematic list, not Texas coordinates or boundaries.

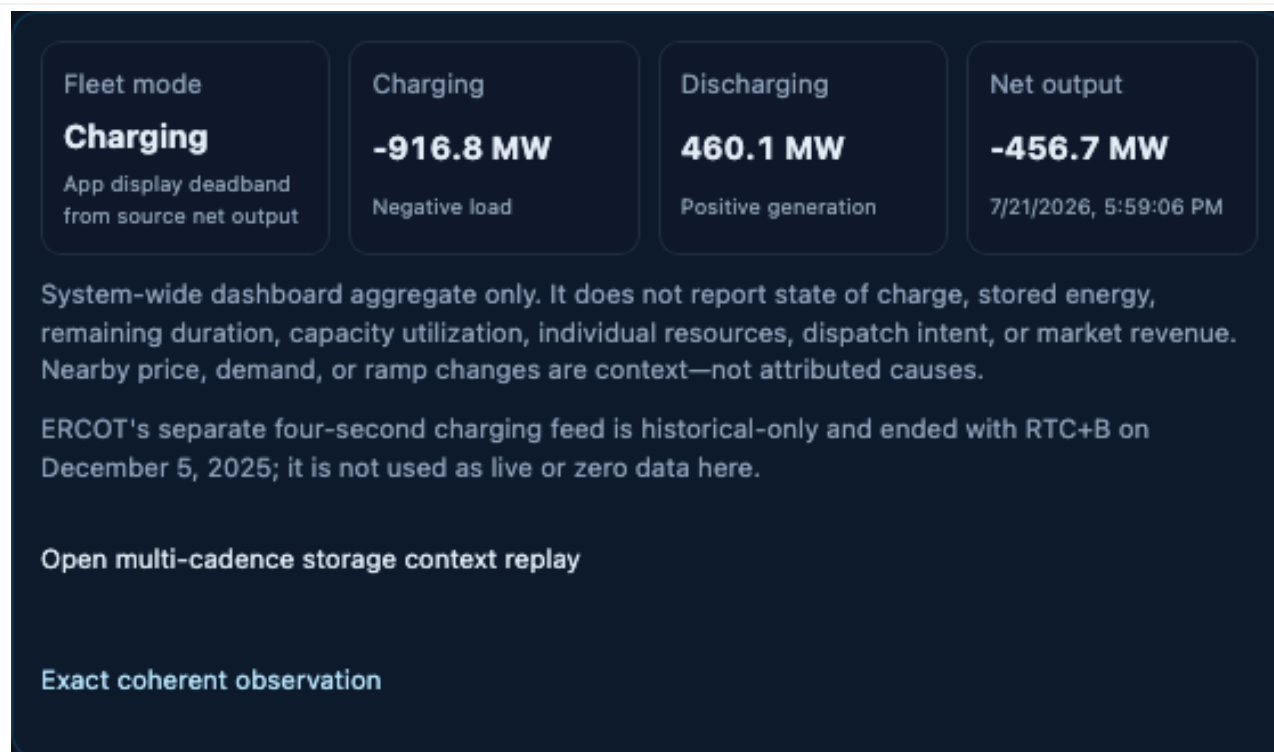
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Houston HU · HB_HOUSTON -\$42.16/MWh	North HU · HB_NORTH \$18.00/MWh	
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Cps LZ · LZ_CPS \$31.00/MWh	Houston LZ · LZ_HOUSTON \$58.00/MWh	
Lcra LZ · LZ_LCRA \$44.00/MWh	North LZ · LZ_NORTH \$92.00/MWh	
Raybn LZ · LZ_RAYBN \$38.00/MWh	South LZ · LZ_SOUTH \$73.00/MWh	
West LZ · LZ_WEST \$225.00/MWh	Busavg SH · HB_BUSAVG \$51.00/MWh	
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HB_PAN	HU	\$35.00/MWh
HB_SOUTH	HU	\$62.00/MWh
HB_WEST	HU	\$145.00/MWh
LZ_AEN	LZ	\$28.00/MWh

17. Storage operations

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium



iPhone WebKit



18. Multi-cadence storage context replay

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

Multi-cadence storage context replay

Recent two-hour source-observation view

Fleet storage, frequency, System Lambda, and available AS capability are shown in the same UTC window. Their different timestamps and cadences are preserved; timing alone does not establish attribution or operational intent.

Alignment: display_window_only · policy:
multi_cadence_context_not_battery_response_attribution

UTC window 2026-07-21T21:00:00.000Z – 2026-07-21T23:00:00.000Z

Storage fleet (MW)



Charging · 300s nominal · Discharging ·
300s nominal · Net output · 300s nominal

System frequency (Hz)



System frequency · 60s nominal

Available AS capability (MW)



Available AS capability · 300s nominal

System Lambda (\$/MWh)



System Lambda · 300s nominal

Source observations and deterministic derived window extrema use separate annotation classes. Official annotations are unavailable because no strict official annotation source is in this slice.

Exact observations and provenance

Derived window annotations

iPhone WebKit

Multi-cadence storage context replay

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Fleet storage, frequency, System Lambda, and available AS capability are shown in the same UTC window. Their different timestamps and cadences are preserved; timing alone does not establish attribution or operational intent.

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UTC window 2026-07-21T21:00:00.000Z – 2026-07-21T23:00:00.000Z

Storage fleet (MW)



Charging · 300s nominal · Discharging · 300s nominal · Net
output · 300s nominal

System frequency (Hz)



System frequency · 60s nominal

Available AS capability (MW)



Available AS capability · 300s nominal

System Lambda (\$/MWh)



System Lambda · 300s nominal

Source observations and deterministic derived window extrema use separate annotation classes. Official annotations are unavailable because no strict official annotation source is in this slice.

Exact observations and provenance

Derived window annotations

19. Predictive weather

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

OPTIONAL NWS CONTEXT

Predictive weather at representative points

Four airport-point forecasts share the Outlook time axis. Temporal overlap is context only and does not establish attribution to an ERCOT load forecast or grid condition.

Hide predictive weather

NWS forecast at representative airport points

Dallas/Fort Worth, Austin, Houston Hobby, and San Antonio are point samples—not ERCOT weather zones, area averages, or statewide extrema.

Dallas/Fort Worth

available

-2.0 °C at Outlook peak

Austin

available

39.0 °C at Outlook peak

Houston Hobby

available

39.0 °C at Outlook peak

San Antonio

available

39.0 °C at Outlook peak

Dallas/Fort Worth exact forecast context

Official NWS grid update 7/21/2026, 5:45:00 PM · retrieved 7/21/2026, 5:46:00 PM

Temperature

-2.0 °C

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Dashboard derived: forecast at or below freezing

Apparent temperature

-5.0 °C

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Heat index

Missing

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind chill

-7.0 °C

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind speed

38.0 km/h

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind gust

65.0 km/h

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Exact NWS forecast intervals

NWS weather alerts

Texas statewide, not ERCOT footprint. NWS weather severity is not an ERCOT grid alert, EEA, or conservation status.

Official NWS · Severe

High Wind Warning for North Texas

Effective 7/21/2026, 5:55:00 PM · expires 7/21/2026, 8:00:00 PM

North Texas counties with a deliberately long area description

Open official NWS alert

Exact NWS alert evidence

Source freshness

iPhone WebKit

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Dallas/Fort Worth

available

-2.0 °C at Outlook peak

Austin

available

39.0 °C at Outlook peak

Houston Hobby

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San Antonio

available

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Dallas/Fort Worth exact forecast context

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Heat index

Missing

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind chill

-7.0 °C

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind speed

38.0 km/h

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Wind gust

65.0 km/h

Exact interval 7/21/2026, 5:00:00 PM–7/29/2026, 5:00:00 PM

Exact NWS forecast intervals

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Exact NWS alert evidence

Source freshness

20. Unified grid event timeline

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

SHARED UTC HISTORY

Unified grid event timeline

ERCOT, NWS, source-observation, and dashboard-derived evidence share a display window. Temporal overlap does not establish attribution or operational intent.

ERCOT operations messages

Collector-accumulated official messages

EEA source observations

Snapshot time is not an official declaration time

NWS weather alerts

Texas statewide, not ERCOT footprint; collected after PR19

TXANS unavailable

No verified authoritative event feed; no events are synthesized

Coverage limits and known gaps

Showing 5 of 5 events, newest first

Evidence All evidence (5)

7/21/2026, 5:06:40 PM

Source observation

eea_source_snapshot

EEA level 2 source snapshot

SOURCE

ercot_daily_prc

EXACT INTERVAL

7/21/2026, 5:06:40 PM

STATUS

2

TIME BASIS

Source snapshot time, not official declaration time

Open official source

Permalink to fixed event window

Open synchronized storage-context window

7/21/2026, 4:50:00 PM

Official ERCOT event

Operational Information

Information

Official ERCOT operations message

Official message body retained independently from weather context.

SOURCE

ercot_operations_messages

EXACT INTERVAL

7/21/2026, 4:50:00 PM

STATUS

Active

TIME BASIS

Exact UTC

Open official source

Permalink to fixed event window

Open synchronized storage-context window

Ambiguous repeated hour - 7/21/2026, 4:16:40 PM or 7/21/2026, 4:33:20 PM

Official ERCOT event

Operational Information

Information

Repeated-hour operations message

Official message body retained independently from weather context.

SOURCE

ercot_operations_messages

EXACT INTERVAL

Ambiguous repeated hour - 7/21/2026, 4:16:40 PM or 7/21/2026, 4:33:20 PM

STATUS

Active

TIME BASIS

Ambiguous repeated Chicago hour

Open official source

Permalink to fixed event window

Replay needs one unambiguous UTC timestamp and a selected window of 24 hours or less

More events are available after this page. Narrow the fixed window to review all exact rows; this page is not presented as complete.

Exact event evidence

iPhone WebKit

SHARED UTC HISTORY

Unified grid event timeline

ERCOT, NWS, source-observation, and dashboard-derived evidence share a display window. Temporal overlap does not establish attribution or operational intent.

ERCOT operations messages

Collector-accumulated official messages

EEA source observations

Snapshot time is not an official declaration time

NWS weather alerts

Texas statewide, not ERCOT footprint; collected after PR19

TXANS unavailable

No verified authoritative event feed; no events are synthesized

Coverage limits and known gaps

Showing 5 of 5 events, newest first

Evidence All evidence (5)

7/21/2026, 5:06:40 PM

Source observation

eea_source_snapshot

EEA level 2 source snapshot

SOURCE

ercot_daily_prc

EXACT INTERVAL

7/21/2026, 5:06:40 PM

STATUS

2

TIME BASIS

Source snapshot time, not official declaration time

Open official source

Permalink to fixed event window

Open synchronized storage-context window

7/21/2026, 4:50:00 PM

Official ERCOT event

Operational Information

Information

Official ERCOT operations message

Official message body retained independently from weather context.

SOURCE

ercot_operations_messages

EXACT INTERVAL

7/21/2026, 4:50:00 PM

STATUS

Active

TIME BASIS

Exact UTC

Open official source

Permalink to fixed event window

Open synchronized storage-context window

Ambiguous repeated hour - 7/21/2026, 4:16:40 PM or 7/21/2026, 4:33:20 PM

Official ERCOT event

Operational Information

Information

Repeated-hour operations message

Official message body retained independently from weather context.

SOURCE

ercot_operations_messages

EXACT INTERVAL

Ambiguous repeated hour - 7/21/2026, 4:16:40 PM or 7/21/2026, 4:33:20 PM

STATUS

Active

TIME BASIS

Ambiguous repeated Chicago hour

Open official source

Permalink to fixed event window

Replay needs one unambiguous UTC timestamp and a selected window of 24 hours or less

More events are available after this page. Narrow the fixed window to review all exact rows; this page is not presented as complete.

Exact event evidence

21. Historical context

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

DASHBOARD OBSERVATIONS

Historical context and records

Close historical context and records

Demand context from observations collected by this dashboard, conditioned on season and America/Chicago civil hour. This is not a forecast or an all-time ERCOT record.

available · The selected completed hour meets the native-observation coverage threshold.

Selected local hour

75.3 GW at Jul 21, 2026, 5:55 PM CDT

2026-07-21 HE 17 · 12 / 12 observations (100%)

Seasonal same-hour range

61.5 GW – 79.1 GW

Type 7 p10–p90 · JJA · 45 qualified prior hours

Completed-day peak rank

3 of 120

2026-07-20 · competition rank · 245 prior days excluded

Exact historical demand evidence and coverage

Evidence	State	Value	Cc
Selected completed civil hour	qualified	75.3 GW at Jul 21, 2026, 5:55 PM CDT	12
Previous local day, same hour	available	73.1 GW at Jul 20, 2026, 5:55 PM CDT	12
Previous local week, same hour	partial	Unavailable	9 /
Previous local year, same hour	unavailable	Unavailable	0 /
Seasonal same-local-hour percentiles	available	p10 61.5 GW; p50 70.3 GW; p90 79.1 GW	Ty

► Method and provenance

iPhone WebKit

DASHBOARD OBSERVATIONS

Historical context and records

Close historical context and records

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Seasonal same-hour range

61.5 GW – 79.1 GW

Type 7 p10–p90 · JJA · 45 qualified prior hours

Completed-day peak rank

3 of 120

2026-07-20 · competition rank · 245 prior days excluded

EvidenceState

Selected completed civil hour

qualified

Previous local day, same hour

availabl

Previous local week, same hour

partial

Previous local year, same hour

unavaili

Seasonal same-local-hour percentiles

availabl

► Method and provenance

22. Texas Grid long-horizon planning

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

Planning snapshots, not operating capacity or realization forecasts

Official monthly reports preserve operational, planned, and studied distinctions. The dashboard does not infer commitments, construction, additions, retirements, or future load.

available

Generator interconnection status

Official study-phase and fuel aggregates. No project rows or identities.

Open interconnection history

available

Long-term load forecast

Official monthly peak MW and energy MWh for two documented forecast scenarios.

Open load forecast

Evidence not yet supported

Large-load project status

Forecast methodology context is available in the LTLF evidence. Individual project categories and status remain unavailable because no stable public machine-readable status dataset is verified.

Gross retirements

Unavailable: no verified gross-retirement source is available.

OFFICIAL ERCOT CAPACITY TREND

Resource capacity trend

Operational capacity and source-defined planned/studied categories remain separate. The official total is displayed as a source total and is never added to or stacked with its component categories. Planning values are not commitments or realization forecasts.

Source period 2026-07 · published Jul 31, 2026, 3:00 PM

Wind

Solar

Battery

Gas - Combined Cycle

Gas - Other

Cadence	Period	Official total	Operational	IA + financial security	IA + no financial security	Other planned
Annual	2024	135 MW	100 MW	20 MW	10 MW	Source column
Planned monthly	2026-08	135 MW	100 MW	20 MW	10 MW	Source column

► Source collection and materialization health

Open official capacity source

iPhone WebKit

Planning snapshots, not operating capacity or realization forecasts

Official monthly reports preserve operational, planned, and studied distinctions. The dashboard does not infer commitments, construction, additions, retirements, or future load.

available

Generator interconnection status

Official study-phase and fuel aggregates. No project rows or identities.

Open interconnection history

available

Resource capacity trend

Official total, operational, planned, studied, and small-generator values.

Close capacity history

available

Long-term load forecast

Official monthly peak MW and energy MWh for two documented forecast scenarios.

Open load forecast

Evidence not yet supported

Large-load project status

Forecast methodology context is available in the LTLF evidence. Individual project categories and status remain unavailable because no stable public machine-readable status dataset is verified.

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Source period 2026-07 · published Jul 31, 2026 at 3:00 PM

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Solar

Battery

Gas - Combined Cycle

Gas - Other

Cadence	Period	Official total
Annual	2024	135 MW
Planned monthly	2026-08	135 MW

► Source collection and materialization health

Open official capacity source

23. External context

Deterministic synthetic review fixture - desktop Chromium and iPhone Pro Max WebKit

Desktop Chromium

iPhone WebKit

External context, not ERCOT operational authority or live emissions

EIA and EPA evidence retains its native cadence, geography, units, and methodology. It never replaces an ERCOT operational feed, and temporal overlap does not establish causality.

Disabled · individual EIA API key not configured

EIA-930 ERCO hourly context

Delayed preliminary hourly energy demand and signed total net interchange.

Open EIA-930 evidence

Disabled · individual EIA API key not configured

Henry Hub daily spot price

Daily source-market-date context without fill, interpolation, or same-hour joins.

Open natural-gas evidence

Available

EPA eGRID annual ERCT rates

Data year 2023, revision 2.

Close eGRID evidence

Unavailable

EPA CAMD hourly emissions

ERCOT-footprint and reporting-coverage methodology is not frozen. This is not a zero-emissions series, and a credential alone cannot enable it.

RETROSPECTIVE ANNUAL EPA METHODOLOGY CONTEXT

eGRID ERCT total-output emission rates

Source-published annual average rates for ERCOT All. They are not current or marginal emissions, generator-specific rates, or ERCOT-wide mass emissions. The dashboard does not multiply these rates by live demand or generation.

Data year 2023 · revision 2 · released 2025-06-12 · ERCT / ERCOT All

Metric ID	Exact source header	Annual average rate	Unit
co2	CO ₂	812.4	lb/MWh
ch4	CH ₄	0.071	lb/MWh
n2o	N ₂ O	0.012	lb/MWh
co2e	CO ₂ e	818.7	lb/MWh
annual_nox	Annual NO _x	0.29	lb/MWh
ozone_season_nox	Ozone Season NO _x	0.25	lb/MWh

► Exact eGRID publication identity

► Source collection and materialization health

Open EPA eGRID source

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Data year 2023, revision 2.

Close eGRID evidence

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RETROSPECTIVE ANNUAL EPA METHODOLOGY CONTEXT

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Data year 2023 · revision 2 · released 2025-06-12 · ERCT / ERCOT All

Metric ID	Exact source header
co2	CO ₂
ch4	CH ₄
n2o	N ₂ O
co2e	CO ₂ e
annual_nox	Annual NO _x
ozone_season_nox	Ozone Season NO _x

► Exact eGRID publication identity

► Source collection and materialization health

Open EPA eGRID source

24. Human review checklist

- ☐ Confirm PR #37 has no generic tile body schema, file write, persistence helper, or exact child route.
- ☐ Confirm fresh-process restart returns the same bytes/ETag without a tile_resources table or artifact file.
- ☐ Confirm corrections change intersecting ranges only and unrelated tiles stay byte-identical.
- ☐ Review the benchmark method and the honestly reported increase in v2 unique URL cardinality.
- ☐ Review all 14 desktop/mobile pairs for containment, source labeling, degraded states, and exact-table access.
- ☐ Confirm official, source-observation, and dashboard-derived provenance remains distinct.
- ☐ Confirm unavailable or blocked source states are never represented as zero.
- ☐ Confirm four-second ESR is accepted as an external scope limit, not silently declared complete.
- ☐ Confirm exact-head CI, range-diff, PR stack topology, secret scan, builds, and Compose evidence.
- ☐ Approve merge separately from deployment; require a second explicit production approval.

Final status

READY FOR DETAILED HUMAN REVIEW / CONDITIONAL MERGE CANDIDATE

Nothing was merged or deployed. Production, Portainer, production SQLite, collectors, Cloudflare, and DNS remain unchanged.

25. Deployment and rollback guide

Execute only after separate merge and production approvals.

Stage	Required action	Stop / rollback signal
Prepare	Record approved revision and matching image digests; export live stack and COMPLETE Env; preserve volumes/network/ports; take coherent DB backup.	Missing rollback artifact or mutable image tag
Initial rollout	Use PullImage=true and Prune=false. Keep every new opt-in disabled. Verify containers, routes, auth, restart counts, freshness, and old behavior.	Restart loop, auth regression, stale operational feeds
Tile smoke	First canonical request MISS; second HIT/304; warm SQLite generation zero. Receiver restart regenerates identical bytes/ETag and no artifacts.	Unexpected persisted body, byte drift, or SQL on warm path
Enable sources	Enable one source at a time. Check collector logs, source timestamps, receiver health, selected pointer, and browser output.	Failure hidden by last-good health or missing source timestamp
Cloudflare	Separate approval. Preserve rule order. Cache only reviewed /api/v2/tiles/* GETs and respect origin revalidation/ETag.	Wrong rule match, cached errors/auth, or missing same-PoP proof
Rollback	Disable collectors first; restore collector then both previous pinned images and COMPLETE prior Env with Prune=false. Restore DB only if required.	Unable to reproduce approved prior runtime

Secret boundary: never place EIA/EPA keys, receiver ingest secrets, credentials, or complete live Env values in screenshots, logs, browser JSON, checkpoints, this packet, or Git. DEMO_KEY is not a production credential.